

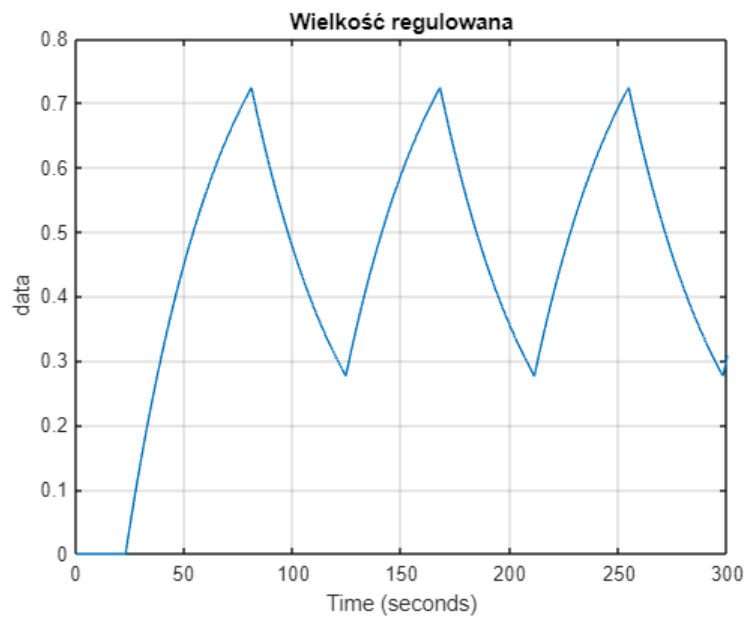
SPRAWOZDANIE

Regulacja dwupołożeniowa

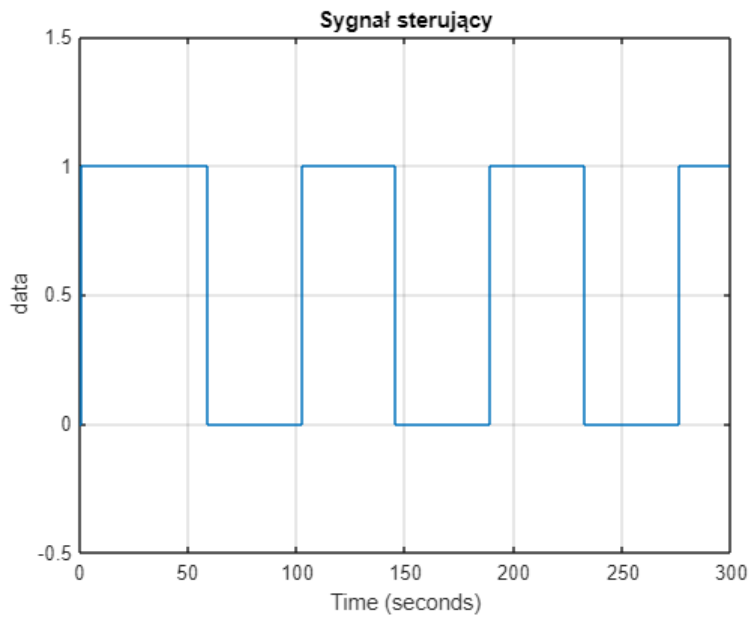
Jan Majerczyk

```
SP = 0.5;  
h = 0.05;  
T = 45;  
tal = 22;  
simOut = sim('sim1_dwupol.slx');  
u = simOut.get("u");  
y = simOut.get("y");
```

```
figure  
plot(y);  
grid on  
title('Wielkość regulowana')
```



```
figure  
plot(u)  
ylim([-0.5 1.5])  
grid on  
title('Sygnał sterujący')
```



```
clear
SP = 0.5;
T = 45;
tal = 22;

h = 0.05;
simOut = sim('sim1_dwupol.slx');
u1 = simOut.get("u");
y1 = simOut.get("y");

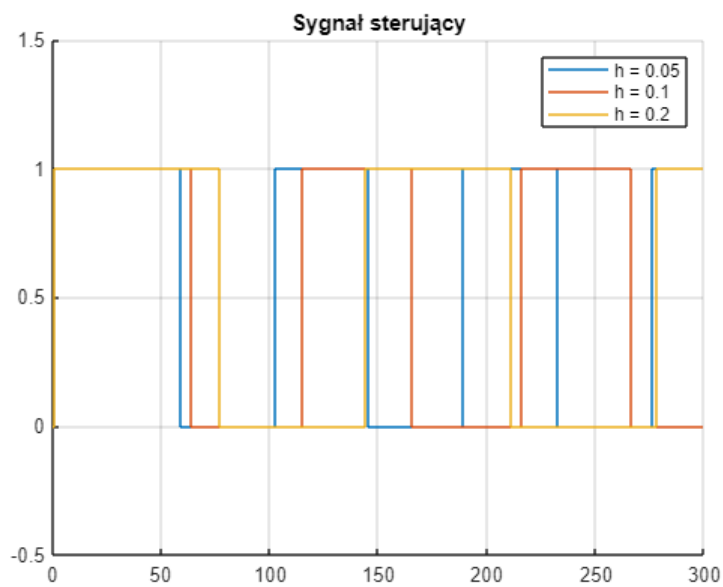
h = 0.1;
simOut = sim('sim1_dwupol.slx');
u2 = simOut.get("u");
y2 = simOut.get("y");

h = 0.2;
simOut = sim('sim1_dwupol.slx');
u3 = simOut.get("u");
y3 = simOut.get("y");

figure
hold on
plot(y1);
plot(y2);
plot(y3);
hold off
grid on
legend('h = 0.05', 'h = 0.1', 'h = 0.2', 'Location','southeast')
title('Wielkość regulowana')
```



```
figure
hold on
plot(u1, '-')
plot(u2, '-')
plot(u3, '-')
ylim([-0.5 1.5])
hold off
grid on
legend('h = 0.05', 'h = 0.1', 'h = 0.2')
title('Sygnał sterujący')
```



```
clear
SP = 0.5;
T = 45;
h = 0.05;
```

```

tal = 10;
simOut = sim('siml_dwupol.slx');
u1 = simOut.get("u");
y1 = simOut.get("y");

tal = 22;
simOut = sim('siml_dwupol.slx');
u2 = simOut.get("u");
y2 = simOut.get("y");

tal = 35;
simOut = sim('siml_dwupol.slx');
u3 = simOut.get("u");
y3 = simOut.get("y");

figure
hold on
plot(y1);
plot(y2)
plot(y3)
hold off
grid on
legend('tal = 10', 'tal = 22', 'tal = 32', 'Location','southeast')
title('Wielkość regulowana')

```



```

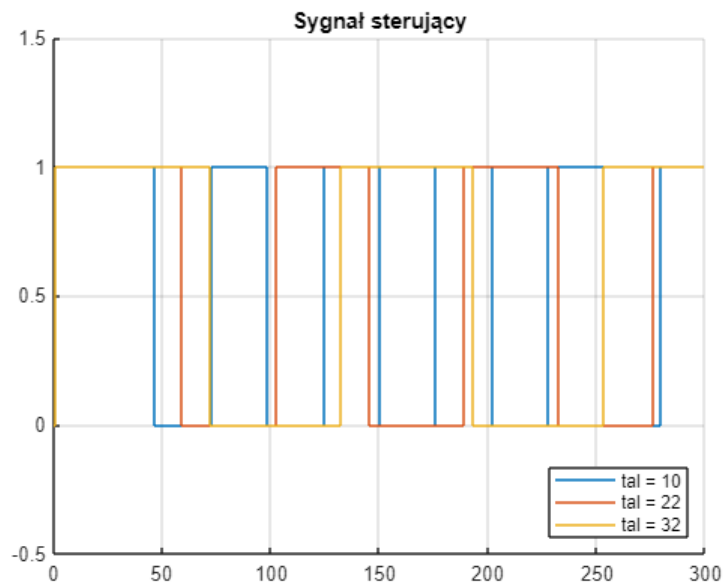
figure
hold on
plot(u1, '- ')
plot(u2, '- ')
plot(u3, '- ')
ylim([-0.5 1.5])
hold off

```

```

grid on
legend('tal = 10', 'tal = 22', 'tal = 32', 'Location','southeast')
title('Sygnał sterujący')

```



```

clear
SP = 0.5;
tal = 22;
h = 0.05;

T = 25;
simOut = sim('sim1_dwupol.slx');
u1 = simOut.get("u");
y1 = simOut.get("y");

T = 45;
simOut = sim('sim1_dwupol.slx');
u2 = simOut.get("u");
y2 = simOut.get("y");

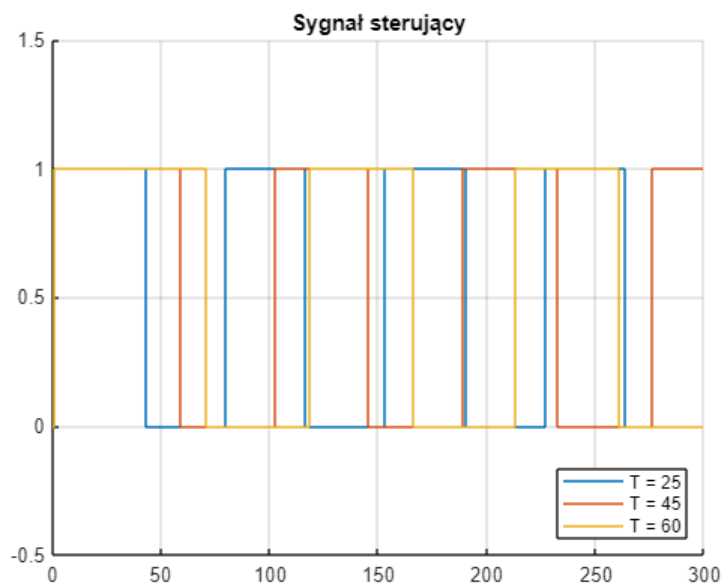
T = 60;
simOut = sim('sim1_dwupol.slx');
u3 = simOut.get("u");
y3 = simOut.get("y");

figure
hold on
plot(y1);
plot(y2);
plot(y3);
hold off
grid on
legend('T = 25', 'T = 45', 'T = 60', 'Location','southeast')
title('Wielkość regulowana')

```



```
figure
hold on
plot(u1, '-')
plot(u2, '-')
plot(u3, '-')
ylim([-0.5 1.5])
hold off
grid on
legend('T = 25', 'T = 45', 'T = 60', 'Location','southeast')
title('Sygnał sterujący')
```



```
clear
tal = 22;
h = 0.05;
T = 45;
```

```

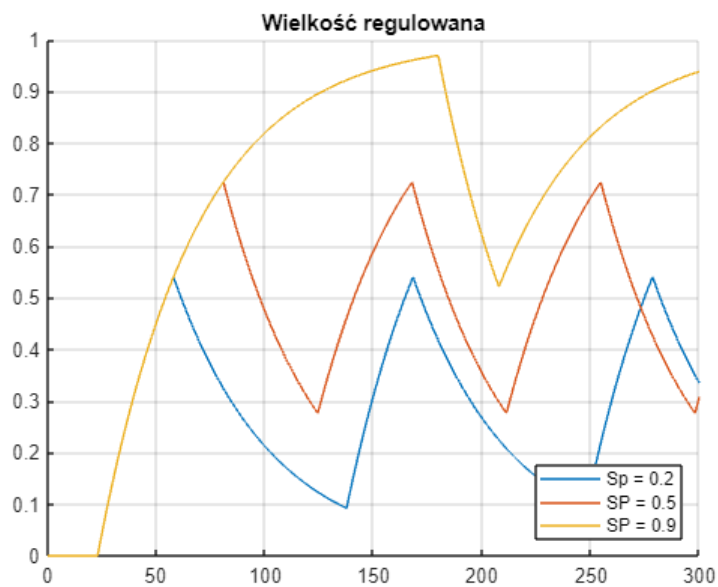
SP = 0.2;
simOut = sim('siml_dwupol.slx');
u1 = simOut.get("u");
y1 = simOut.get("y");

SP = 0.5;
simOut = sim('siml_dwupol.slx');
u2 = simOut.get("u");
y2 = simOut.get("y");

SP = 0.9;
simOut = sim('siml_dwupol.slx');
u3 = simOut.get("u");
y3 = simOut.get("y");

figure
hold on
plot(y1);
plot(y2);
plot(y3);
hold off
grid on
legend('Sp = 0.2', 'SP = 0.5', 'SP = 0.9', 'Location','southeast')
title('Wielkość regulowana')

```



```

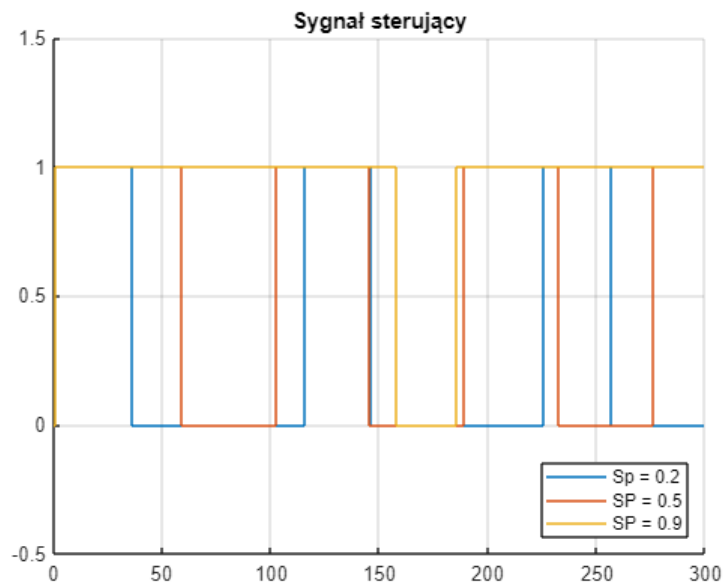
figure
hold on
plot(u1, '- ')
plot(u2, '- ')
plot(u3, '- ')
ylim([-0.5 1.5])
hold off

```

```

grid on
legend('Sp = 0.2', 'SP = 0.5', 'SP = 0.9', 'Location','southeast')
title('Sygnał sterujący')

```



```

clear
tal = 22;
h = 0.05;
T = 45;

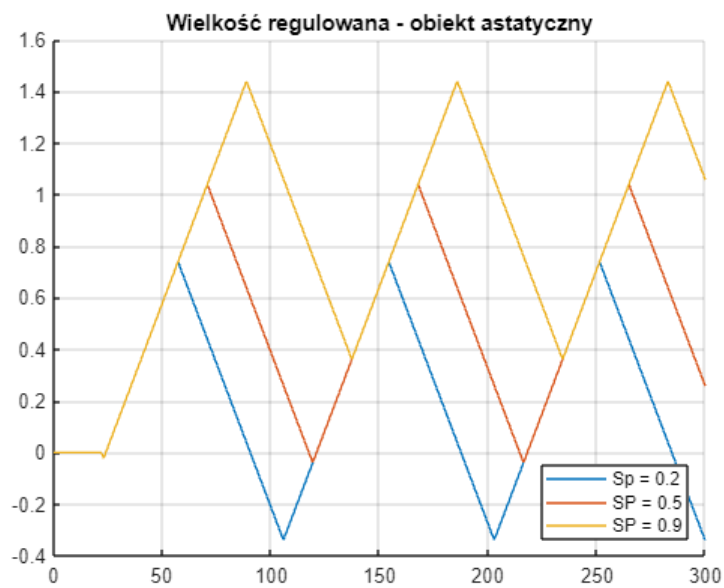
SP = 0.2;
simOut = sim('sim1_dwupol.slx');
u1 = simOut.get("au");
y1 = simOut.get("ay");

SP = 0.5;
simOut = sim('sim1_dwupol.slx');
u2 = simOut.get("au");
y2 = simOut.get("ay");

SP = 0.9;
simOut = sim('sim1_dwupol.slx');
u3 = simOut.get("au");
y3 = simOut.get("ay");

figure
hold on
plot(y1);
plot(y2);
plot(y3);
hold off
grid on
legend('Sp = 0.2', 'SP = 0.5', 'SP = 0.9', 'Location','southeast')
title('Wielkość regulowana - obiekt astatyczny')

```

```
figure
hold on
plot(u1, '-')
plot(u2, '-')
plot(u3, '-')
ylim([-0.5 1.5])
hold off
grid on
legend('Sp = 0.2', 'SP = 0.5', 'SP = 0.9', 'Location','northeast')
title('Sygnał sterujący - obiekt astatyczny')
```

